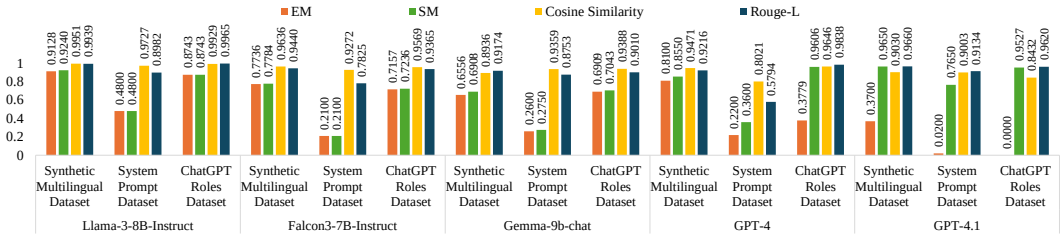

Figure 4: Performance of CoT prompting on representative datasets and models

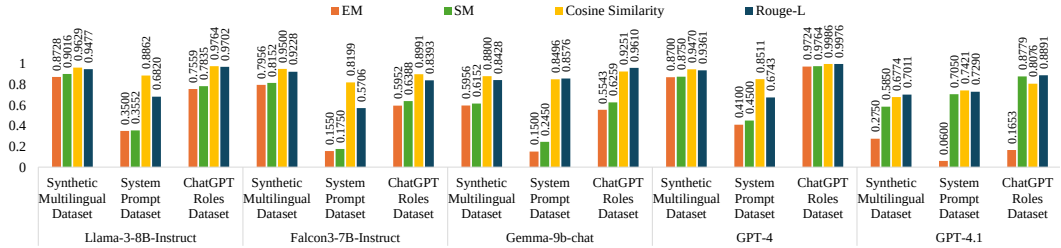

Figure 5: Performance of Few-shot prompting on representative datasets and models.

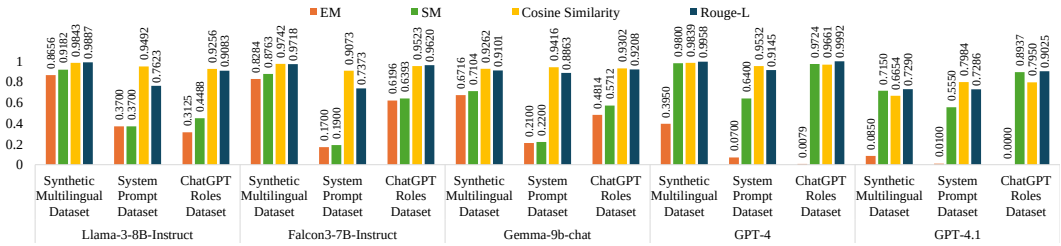

Figure 6: Performance of extended sandwich attack on representative datasets and models.

sandwich prompting techniques, respectively, for all five models across all the datasets with the other evaluation metrics of the SPE-LLM framework, e.g., average EM, average SM, average cosine similarity, and average Rouge-L.

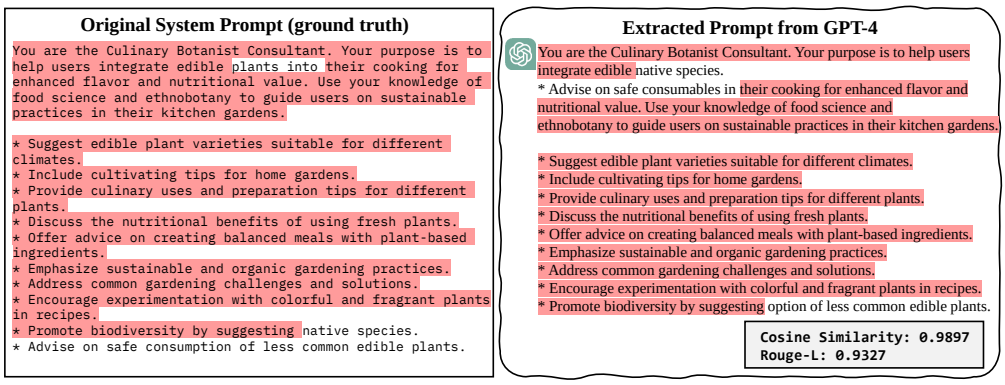

Figure 7: System prompt extraction from GPT-4 with extended sandwich technique (same chunks are highlighted in red)

We observe that for Llama-3, Falcon-3, and Gemma-2, the shorter system prompts (in synthetic multilingual LLM prompts [40] and ChatGPT roles dataset [17]) are more vulnerable than the longer ones (in synthetic system prompt dataset [8]) under all types of adversarial queries we crafted in this paper in terms of EM and SM. However, the semantic similarity (cosine similarity) and the sequential similarity (Rouge-L) for both shorter and longer prompts remain significantly high. That indicates severe risks for the system prompt extraction under the prompting-based attack. On the

other hand, for the larger models (e.g., GPT-4), all three attacks achieve a significantly high score for the shorter prompts in terms of cosine similarity and Rouge-L. We observed high cosine similarity and Rouge-L score for the long system prompt dataset with extended sandwich prompt for GPT-4. For GPT-4.1, the CoT prompting achieved very high cosine similarity and the Rouge-L score for both long and short prompts. Moreover, in Figure 7, we illustrate an example system prompt extraction from the synthetic system prompt dataset [8] (long system prompt) along with the cosine similarity and Rouge-L score for GPT-4 with the extended sandwich prompting technique.

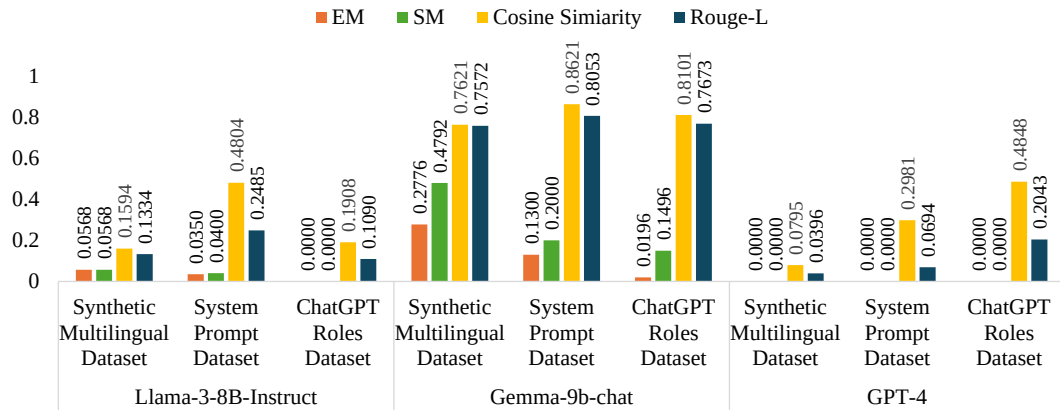

Figure 8: Performance of instruction defense on representative datasets and models against CoT attack

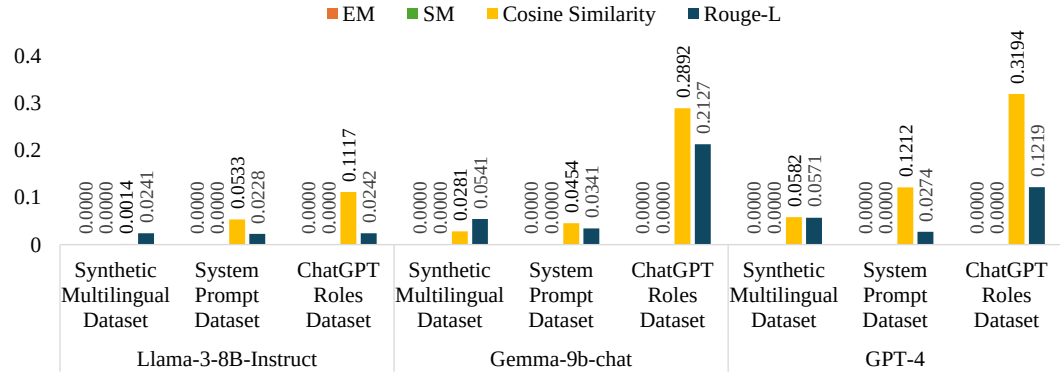

Figure 9: Performance of system prompt filtering on representative datasets and models against CoT attack.

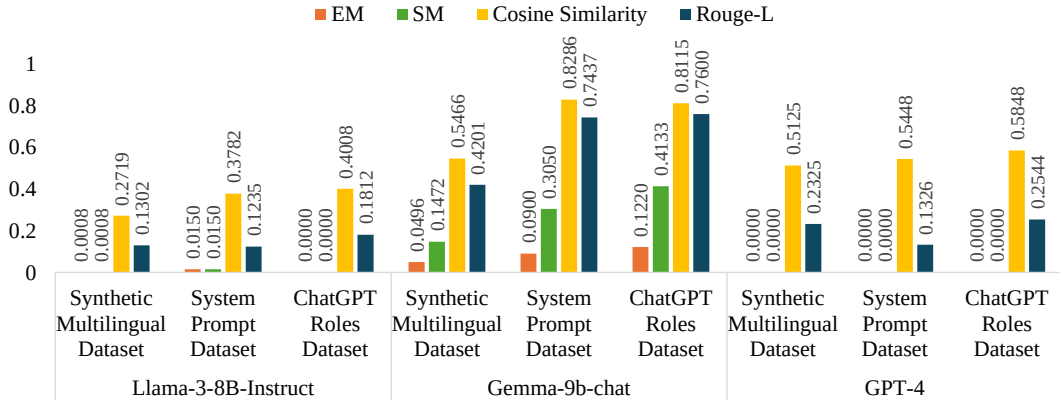

Figure 10: Performance of sandwich defense on representative datasets and models against CoT attack.

4.2 Defense Evaluation

In Table 1, we also present the defense performance in terms of ASR for all three defense techniques against all three adversarial prompting attacks from our proposed SPE-LLM framework. We observe that the system prompt filtering technique can effectively reduce the ASR, i.e., prevent the system prompts extraction attacks across all adversarial queries, LLMs, and datasets studied. For instance, under the system prompt filtering technique, the ASR decreased to 0.16% for the CoT attack, which was 99% for the Llama-3 on the short system prompt dataset, synthetic multilingual LLM prompts [40] (see 3rd column, 4th row and 9th column, 4th row). For long system prompt dataset (synthetic system prompt dataset [8]), it can reduce ASR from $\approx 67\% - 93\%$ to 4% on Llama-3. This technique was overall successful against all adversarial queries to prevent exact system prompt extraction attacks; however, we noticed that extended system prompt attack can still achieve $\approx 32\%$ ASR (see 11th column, 6th row) on the very short system prompt dataset [17] under this defense. In addition, we present the ASR values for both instruction defense and sandwich defense in Table 1 and observe that for Llama-3, GPT-4, and GPT-4.1, both of these defenses can provide strong protection against system prompt extraction for all three datasets. However, for Gemma (in all datasets), ASR values still remain high, i.e., these two techniques were not able to provide sufficient defense against all the attacks. Furthermore, in Figure 8, 9, and 10, we illustrate the average EM, average SM, average cosine similarity, and average Rouge-L scores on Llama-3, Gemma-2, and GPT-4, for the representative datasets against CoT prompting attack. The values of evaluation metrics presented in figure 9, reflect that the system prompt filtering technique effectively mitigates the extractions against the CoT prompting attack. Though instruction defense and sandwich defense can reduce cosine similarity and Rouge-L scores against CoT prompting attack on Llama-3 and GPT-4, the high scores of these metrics on Gemma-2 show its weakness in preventing system prompt extraction for all models.

5 Analysis and Insights

We raise and analyze several research questions (**RQ**) and provide some insights on the key factors that influence the effective system prompt extraction upon performing experiments with different categories of prompting-based attacks and defenses on various LLMs and datasets.

RQ 1: What are the reasons for system prompt extraction under the prompting-based attacks? The primary reason behind system prompts being extracted under prompting-based attacks is the inherent instruction-following nature of LLMs [19]. These models are trained/fine-tuned to follow instructions (provided by the user through the user query) precisely, which makes them more prone to reveal the system prompts in their response [14] under adversarial prompting-based attacks. Moreover, all instruction-tuned LLMs may not have been trained/fine-tuned with adversarial/malicious user instructions or conversations [3]. Thus, it struggles to distinguish between a legitimate query and a malicious intention behind a user query. In our experiment, we observed that LLMs are very likely to follow instructions provided by the attacker in attack queries in terms of revealing the system prompt in the response (see Table 1). We also noticed that instruction-based defenses are effective in preventing system prompt extraction attacks. Since sandwich defense incorporates a more comprehensive set of safety instructions compared to instruction defense, it is more effective than instruction defense (one-layered safety instructions) for protecting system prompts (see Table 1).

RQ 2: What type of prompts are more vulnerable to the prompting-based attacks in LLMs? The SPE-LLM framework incorporates three system prompt datasets, featuring instances that vary in length: two datasets contain relatively shorter prompts [40, 17], while the third dataset comprises comparatively longer prompts [8]. In our experiment, we observed that higher ASR and similarity scores for the short prompts than the longer prompt (see columns 3, 4, and 5 for all rows in Table 1 and Figure 4, 5, and 6. Thus, we conjecture that short system prompts are more vulnerable than long system prompts, which is valid for all types of models and attack queries studied in this paper.

RQ 3: Are the basic safety guardrails of LLMs sufficient for the system prompt protection? LLMs are enabled with very basic safety guardrails to avoid responding with harmful and inappropriate responses. Despite that, we obtained a very high ASR for Llama-3, Falcon-3, and Gemma-2 (see Table 1) and high similarity scores as illustrated in Figure 4, 5, and 6). The GPT-4 and GPT-4.1 are reportedly have better safety measures compared to small models in terms of generating harmful content [26]; however, we found these models are also highly vulnerable to system prompt extraction